

FERTILIX ANALYTICS - IVF SUCCESS FACTORS EXPLORATION PLATFORM

1. INTRODUCTION

This project demonstrates a complete **end-to-end data analysis workflow** using Python and Power BI.

The main objective is to:

- Understand the dataset
- Clean and preprocess data
- Perform numerical/statistical analysis
- Visualize insights
- Build an interactive dashboard

Tools Covered:

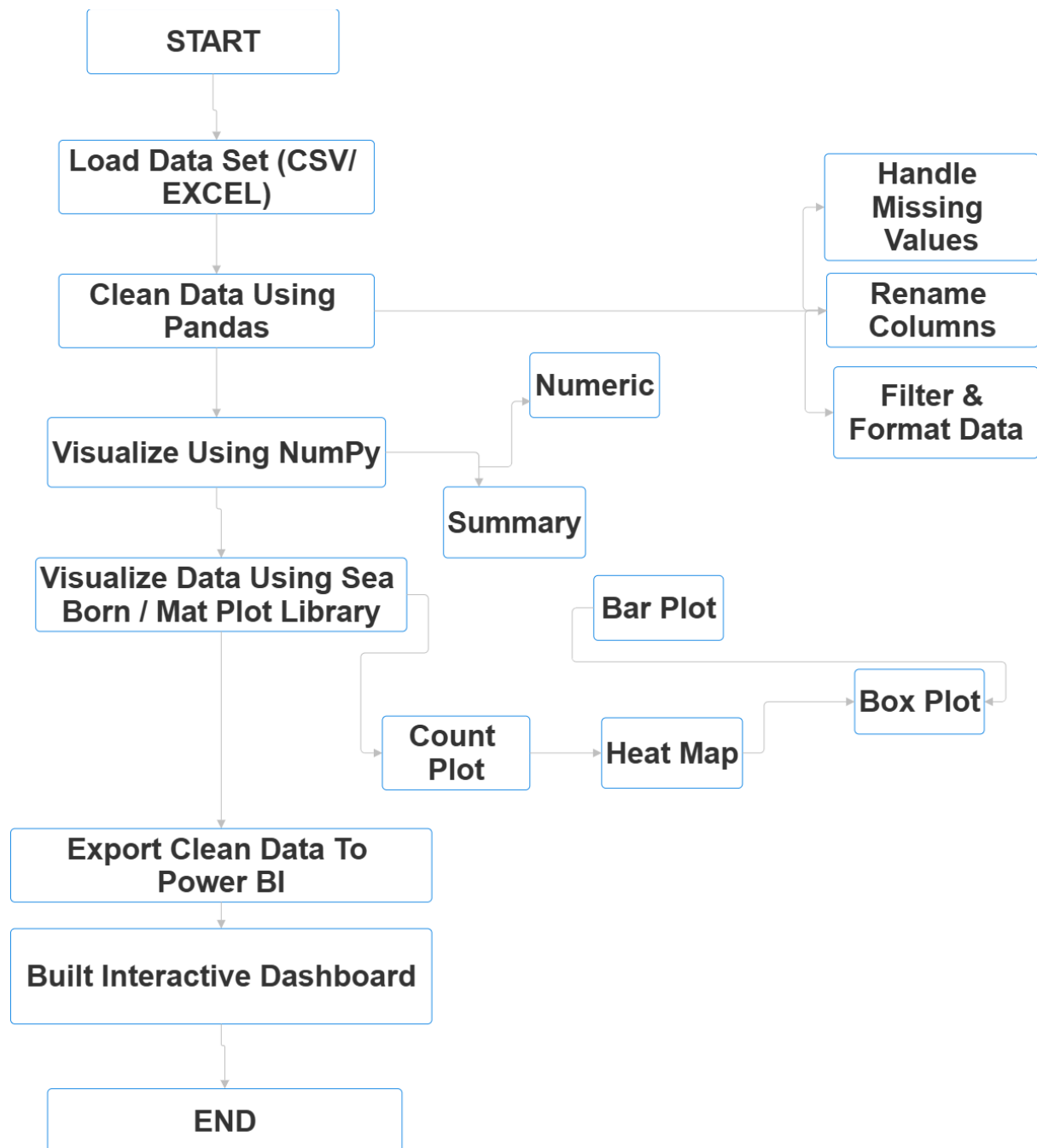
- **NumPy** → Numerical computations
- **Pandas** → Data cleaning & handling
- **Seaborn** → Statistical visualizations
- **Matplotlib** → General plotting
- **Power BI**: Dashboard creation

2. WORKING

- **Import Python libraries**
- **Load dataset using Pandas**
- **Explore dataset (shape, types, missing values)**
- **Clean & preprocess data**
- **Perform numerical/statistical operations (NumPy)**
- **Visualize patterns and relationships (Seaborn, Matplotlib)**

- **Export final dataset to Power BI**
- **Create dashboard with charts & KPIs**
- **Interpret results**

3.FLOWCHART



4. PYTHON PROGRAM

```
import NumPy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load data
df = pd.read_csv("dataset.csv")

# Explore
df.info()
df.describe()

# Visualization examples
sns.countplot(x="Gender", data=df)
plt.show()
```

5. VISUALIZATION METHODS

Using Matplotlib

- Bar graph
- Line graph
- Histogram
- Pie chart
- Scatter plot

Using Seaborn

- Count plot
- Heatmap
- Boxplot
- Pair plot

6. POWER BI STEPS

- Import cleaned dataset (CSV)
- Transform data (Power Query if needed)
- Create:
 - Bar charts
 - Pie charts
 - Card visuals
 - Slicers
- Build an **interactive dashboard**
- Highlight key findings

7. SUMMARY OF PROJECT

- Loaded and cleaned dataset using Pandas
- Performed numerical analysis using NumPy
- Created meaningful visualizations using Seaborn/Matplotlib
- Designed a complete dashboard in Power BI
- Understood relations between different dataset features

8. WHAT I HAVE ANALYZED THE DATA

- How to read & clean data using Pandas
- How to perform calculations using NumPy
- How to make visualizations
- How to interpret graphs
- How to build dashboards in Power BI
- How to follow an end-to-end data science workflow.

9. Results

- Loaded and explored the IVF dataset using **Pandas**

- Cleaned and pre-processed data (handled missing values, renamed columns, formatted data)
- Performed **numerical/statistical analysis** with NumPy (summary statistics, correlations)
- Created **visualizations** with Seaborn and Matplotlib (count plots, heatmaps, boxplots, scatter plots)
- Exported the cleaned dataset into **Power BI**
- Built an **interactive dashboard** with charts, KPIs, and slicers
- Generated insights about IVF success factors and relationships between dataset features